5 A box contains small dust particles suspended in air.

(a) The dust particles move in a random motion.

Complete the sentences to explain the motion of the dust particles.

Air molecules inside the box are very small and move freely. The air molecules also move ..... and ..... .

Air molecules hit dust particles causing the dust particles to ..... .

[3]

(b) The temperature of the air inside the box increases.

State and explain, in terms of particles, the effect on the air pressure inside the box.

statement .....

explanation .....

.....

.....

.....

[3]

(c) The air temperature inside the box is 296 K.

Calculate the air temperature in °C.

temperature = ... °C [1]

[Total: 7]

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